On this account of hypothesis testing, the investigator is intended to set the significance of the test as a function of his or her epistemic attitude towards the null hypothesis, resulting in an asymmetry between the null and alternative hypothesis. This epistemic attitude is *not* to be confused with the notion of a prior probability assigned to the hypothesis, but rather with the investigator's tolerance for a Type I error (i.e. the probability that the test mistakenly rejects the null hypothesis).

Many introductory texts in statistics give an interpretation of hypothesis testing in this manner: one *retains* H_0 in case $p > \alpha$. Wasserman [13], for instance, gives this interpretation, likening hypothesis testing to a criminal trial:

> [H]ypothesis testing is like a legal trial. We assume someone is innocent unless the evidence strongly suggests that he is guilty. Similarly, we retain H_0 unless there is strong evidence to reject H_0. (Page 150)

In this telling, the principle of "presumption of innocence until proven guilty" plays the same formal role as "retaining the null unless it is rejected with significance α." Such principles are *default* principles in the sense that they normatively constrain one's beliefs or actions to a default position (acquittal of the defendant in the case of a criminal trial and retention of the null in the case of hypothesis testing). For example, the presumption of innocence can be interpreted as constraining a factfinder against convicting a defendant in the special case that the prosecution provides *no* evidence of the guilt of the defendant, regardless of the factfinder's underlying personal credence regarding the defendant's guilt.

There are two major problems with the default account of hypothesis testing that render it implausible. First, and most simply, the investigator need not render a decision in favor of either H_0 or H_1, by contrast with the decision problem posed in a criminal trial. One may have *intermediate attitudes* toward the hypothesis H_0 both prior to and following the experiment. For example, one could for example *affirm* H_0, be *ambivalent to* H_0, or *reject* H_0 in response to evidence E. Such a system would require at minimum three attitude towards a hypothesis, more than the two given in the orthodox account.

Secondly, simply *declaring* H_0 to be the null hypothesis does not *warrant* its being taken as default. This issue is most salient when considering the notion of Type II errors, which occurs when a test fails to reject a false hypothesis. On the default view of hypothesis testing, failing to reject H_0 results in accepting H_0. By designating H_0 as the default, a Type II error *is* a genuine error in belief. But it is an error only if one accepts the normative force of the inference to the default in case of a failed test. Implicit in referring to a Type II error as an *error* is an underlying model of default logic as *the* logic of null hypothesis statistical testing.

To avoid such errors therefore requires a theory of warrant for a default hypothesis. In practical settings, however, warrant is typically not given. It is standard practice to test the two-sided null hypothesis of "no difference" even when the investigator has reason to believe that this hypothesis is false. Consider the two-sided null hypothesis is $H_0 : \theta = 0$, where $\theta = \mu_A - \mu_B$ is the difference in mean between two populations. Typically, justification for H_0 is not given, and H_0 may itself be implausible at the outset of the investigation. For example, suppose that I already have knowledge that the quantity μ_i is causally affected by some property P, and that the base rate of P between groups A and B differs substantially. Then the null hypothesis H_0 can be plausibly doubted along these lines, so is not a reasonable default position to take, *despite* its apparent symmetry between A and B.

This is not to say that there are no situations in which the default account of hypothesis testing may be of value. The default account does provide a discursive framework by which claims made and believed by an interlocutor can be treated as default and subjected to scrutiny through statistical analysis. However, this discursive paradigm does not exhaust the modes of scientific discourse, which may for example include scenarios where the investigator has no prior attitude toward either H_0 or H_1 and would like to assess whether *either* of them has strong evidence in its favor.

Through its reliance on the specification of a default hypothesis and the bivalence of the testing procedure, the default account of hypothesis testing fails to provide a framework that adequately models the various epistemic attitudes an investigator might reasonably have towards a hypothesis.

2.2. **The Falsificationist Account of Hypothesis Testing and Frick's Counterargument.** Owing to problems in the interpretation of p-values, and motivated by Popper's falsificational account of scientific theories, some maintain that the only normative role that hypothesis testing plays is that of refuting the null hypothesis. This account stands in contrast to the default account in terms of the admissible values for a test: either H_0 is rejected, or H_0 is not rejected. Thus, a Type II error is no longer an error in the investigator's updated belief in H_0. But, like the default account of hypothesis testing, there is an asymmetric role between the null and alternative hypotheses and the test procedure is bivalent.

In a highly cited paper, Frick reconstructs an argument as to *why* people might think the null hypothesis ought never be accepted. He writes:

> In any given interval (e.g., from -10 to +10), there are an infinite number of real numbers. When the answer to a question could be any real number in an interval, and when the probability of each real number being the answer is approximately the same, the finite amount of probability must be spread over the infinite number of real numbers. Each individual real number must receive essentially zero (i.e., infinitesimal) probability, and only an interval of real numbers can receive a nonzero probability. (A nonzero probability is one greater than essentially zero; winning the lottery, for example, has a nonzero probability.)
>
> This logic would be applied (incorrectly) to the problem of accepting the null hypothesis in the following way: The question concerns the size of the effect one variable has on another, with size zero corresponding to the null hypothesis. Zero is just one point in the interval of infinite possible answers, so the probability of the effect being exactly 0.00000... (with the zeros being carried out to an infinite number of decimal places) is essentially zero. Therefore, the null hypothesis is impossible. (Pages 132-133)

It is worth highlighting that only two-sided hypotheses are contained within the scope of this argument, since hypotheses of the form $H_0 : \theta \in [L, U]$ with $L < U$ can have positive measure. From a frequentist perspective, the hypothetical argument that Frick describes is indeed untenable. Frequentists do not recognize the meaningfulness of a nontrivial *probability* measure on the space of hypotheses.

To see why, let us first formalize the argument given above in a Bayesian paradigm. Let $\Theta = [0, 1]$ and let $H_0 : \theta = \frac{1}{2}$. Then with respect to the uniform prior, or indeed any atomless prior, on Θ, $\mathbb{P}(H_0) = 0$. Thus $\mathbb{P}(H_0|E) = 0$ for any evidence accrued, and so H_0 is unconfirmable even if true. Unfortunately, this argument is not admissible within a frequentist paradigm.

For a frequentist, probability statements of the form $\mathbb{P}(H)$ can be formally defined, but are not epistemically useful. Let $\theta_\top$ be the true value of the parameter θ. Given $\theta_\top$, one can assign the probability that H is true to be

$$\mathbb{P}_\top(H) = \begin{cases} 1 & \text{if } \theta_\top \in H; \\ 0 & \text{if } \theta_\top \notin H. \end{cases}$$

$\mathbb{P}_\top$ assigns a probability of H equal to its truth value: H has positive probability if and only if H has probability 1 if and only if $\theta_\top \in H$. With respect to this measure, $\mathbb{P}(\{\theta_\top\}) = 1$ and so $\mathbb{P}(\{\theta_\top\}|E) = 1$. Thus, there *exists* a probability measure for which $\theta_\top$ is correctly confirmed conditional on any evidence. At first glance, this may suggest that $\{\theta_\top\}$ is confirmable. But this is incorrect. One cannot know the probability assignments of $\mathbb{P}_\top$ *a priori*: exhaustive knowledge of the probability assignments of $\mathbb{P}_\top$ would entail knowledge of the true value $\theta_\top$.

It would seem therefore that arguing that two-sided hypotheses $H_0 : \{\theta\}$ are unconfirmable does not fit cleanly within a measure-theoretic framework for the frequentist.

Frick's argument in favor of the confirmability of the null relies on a probability theoretic grounding inadmissible to the frequentist:

> However, the argument above does not necessarily apply to accepting the null hypothesis. A few special numbers can be allocated a nonzero probability before spreading the remaining probability over the remaining real numbers. In the scales used by psychologists, zero is usually a special, meaningful number. Therefore, the number zero can be assigned a nonzero probability of occurring without violating any rules of probability. (Page 133)

Frick correctly rebuts the Bayesian argument given above: for a Bayesian investigator it is absolutely correct that "the number zero can be assigned a nonzero probability of occurring without violating any rules of probability," and that one can "allocat[e] a nonzero probability" to the value of the effect size being exactly zero. These considerations suggest that one may, in certain circumstances where there is expert knowledge, be justified in having a prior probability of truth assigned to a two-sided hypothesis greater than 0, and therefore be confirmable in the Bayesian sense as the conditional probability $\mathbb{P}(H_0|E)$ has the possibility of becoming arbitrarily close to 1.

Despite the apparent success of Frick's argument, it ultimately rests on the assignment of prior probabilities to hypotheses inadmissible to the frequentist. In frequentist inference, there is no sense in talking about a prior *probability* of the hypothesis in question.

In the following section I provide an alternative, frequentist analysis of the possibility of confirming a two-sided hypothesis. I will argue that the conclusion Frick responds to is correct: two-sided hypotheses in continuous parameter spaces cannot be confirmed. This is not due to any measure theoretic consideration, but instead is due to the *topological* defects of two-sided hypotheses. The argument I will give formalizes Lakens' claim that "it is statistically impossible to support the hypothesis that a true effect size is exactly zero." [5].

2.3. A Middle Path: Confidence Intervals, Decision, and Indecision.

2.3.1. *Duality Between Confidence Intervals and Hypothesis Testing.* Recall from [4] (Definition 9.17) that a two-sided $1-\alpha$ confidence interval (or, more generally, a *confidence region*) is a random set $c(E)$ which is a function of the collected data (or "evidence") E such that for each $\theta \in \Theta$

$$\mathbb{P}_\theta(\theta \in c(E)) \geq 1-\alpha. \tag{1}$$

I emphasize that this is *not* a probability statement about θ; rather, it is a probability statement about the properties of the stochastically-generated object $c(E)$ [13].[1]

For many common tests, rejection of H_0 at level α is equivalent to a property of a $1-\alpha$ confidence interval $c(E)$:

$$\text{Reject } H_0 \Leftrightarrow c(E) \subseteq H_0^c \tag{2}$$

In other words, one rejects H_0 at the α level in case the associated confidence interval $c(E)$, a nonempty open subset of Θ, is fully contained within the complement of H_0. This apparent reduction of hypothesis testing to set-theoretic relationships between confidence intervals and subsets of the underlying parameter space render it amenable to modal interpretation.[2]

By contrast with the previously discussed interpretations of hypothesis testing, the utilization of confidence intervals allows us to remedy the epistemic issues discussed above.

1. (Symmetric Role of the Null and Alternative Hypotheses) The null and alternative hypotheses H_0 and H_1, neither of which will be accepted at default.
2. (Multivalent Decision Problem) There are three potential values for the decision problem for each hypothesis H_i:

[1] The dependence of the event $\theta \in c(E)$ on the stochastic nature of $c(\cdot)$ can be readily seen by writing the event as $X_\theta = \{E \in \Omega^n \mid \theta \in c(E)\}$, which is a subset of Ω^n.

[2] An explanation of the duality between estimation of the confidence interval and hypothesis testing can be found in [4] section 12.4: "Duality between testing and interval estimation."